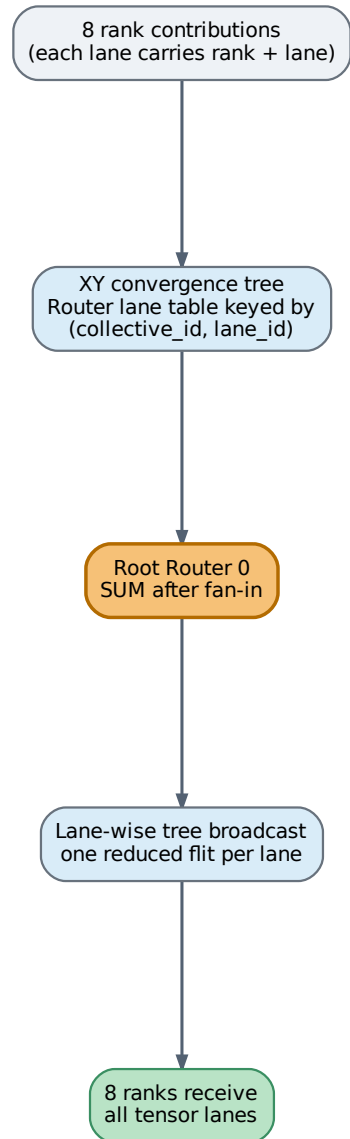


Tensor all-reduce datapath



Scalar lowering

lane 0

lane 1

...

lane 6,719

6,720 logical requests
serialized scalar lanes

one flit = one 64-bit reduction lane

Tensor lowering

HEAD

BODY × 62

TAIL

15 logical requests × 64 flits
lane_id remains attached to each flit

same network work
53,760 contribution flits
147,840 Router flits

measurement window
scalar 80.638 M ticks → tensor 17.055 M ticks
4.728× shorter